

Quiz (Habanero)

- Q1.** The graph of $y = |x - 2|$ has a vertex at
(A) (0, 2) (B) (2, 0) (C) (2, 2) (D) (0, 0) (E) (1, 1)
- Q2.** What is the axis of symmetry for the graph of $y = |x + 1|$?
(A) $x = 0$ (B) $x = 1$ (C) $x = -1$ (D) $x = 2$ (E) $x = -2$
- Q3.** The direction of opening for the graph of $y = |x|$ is
(A) Up (B) Down (C) Left (D) Right (E) None
- Q4.** The graph of $y = 2|x - 3|$ has a vertex at
(A) (3, 0) (B) (3, 2) (C) (2, 3) (D) (0, 3) (E) (1, 1)
- Q5.** If the temperature in a region is given by $y = |x - 10| + 20$, where x is the number of hours since sunrise, what is the temperature at sunrise?
- (A) 20 (B) 30 (C) 10 (D) 0 (E) 40
- Q6.** A rain gauge measures the amount of rainfall in a region. If the amount of rainfall is given by $y = |x| + 2$, where x is the number of days since the start of the month, what is the amount of rainfall on the 5th day of the month?
(A) 5 (B) 7 (C) 3 (D) 2 (E) 9
- Q7.** The ecosystem of a forest is affected by the temperature, which is given by $y = |x - 5| + 10$, where x is the number of months since January. What is the temperature in June?
(A) 10 (B) 15 (C) 20 (D) 5 (E) 25
- Q8.** The graph of $y = |x| - 2$ has a minimum value of
(A) -2 (B) 0 (C) 2 (D) -1 (E) 1

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = |x + 2| - 3$ and identify the vertex, axis of symmetry, and direction of opening. Provide a brief explanation of each.
- Q10.** The water level in a lake is affected by the amount of rainfall, which is given by $y = |x - 4| + 1$, where x is the number of weeks since the start of the year. If the lake is currently at a water level of 5, what is the amount of rainfall that has occurred so far this year? Provide a step-by-step solution and explain your reasoning.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the properties of absolute value functions.
- Tip 2: Encourage students to use graphing tools to visualize the functions.
- Tip 3: Provide additional support for students who struggle with transformations and graphing.